Scalar-local 31+32τ digit synthesis: cold path and full replay

Question. Can the k=5 secp256k1 Horner path build only the digit points used by one scalar, and what does this change for one cold scalar versus a full frozen stream?

Scope and inputs. This experiment uses SEC 2 secp256k1, $E: y^2 = x^3 + 7$ over $p = 2^{256} - 2^{32} - 977$, with subgroup order $n = \texttt{FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFEBAAEDCE6AF48A03BBFD25E8CD0364141}$. The curve catalog has no EC1 secp256k1 record; these exact parameters identify the instance. The full fixture has 12,288 distinct scalar/point pairs and SHA-256 a1c4964f174e36ea26f6a1897128f651a2ce7251e4bf4ffd532ff09221e721d3.

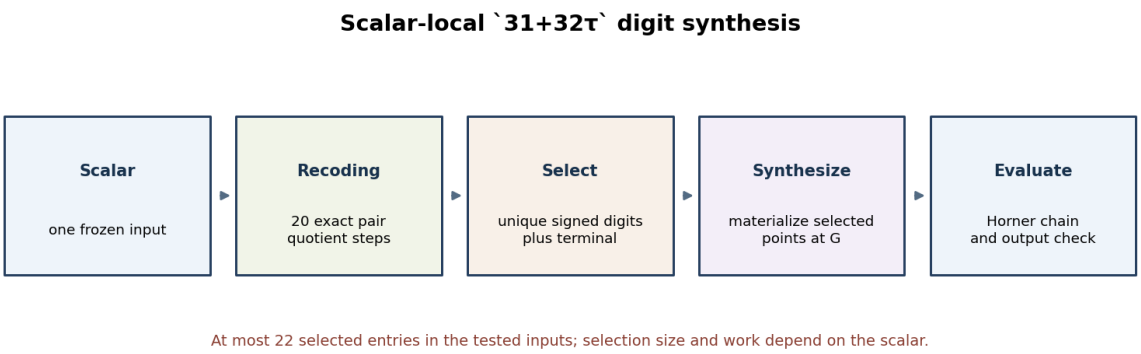


Figure 1: Scalar-local digit synthesis pipeline

Implementation review. The hash-guarded native overlay starts with the existing signed-pair recoder, gathers all twenty 31+32τ digits, adds the terminal pair and a zero slot, then materializes the distinct selected points at G. It evaluates the same Horner chain and checks outputs against the fixture. The two frozen inputs used at most 22 selected entries, including the zero slot. This is an entry-count observation, not a peak-memory measurement; temporary arrays and maps also allocate.

The selected set and path length depend on the scalar: digit pairs, deduplication through BTreeSet/BTreeMap, point synthesis, terminal identity handling, and zero-digit additions vary with digit values. The implementation is therefore not constant-time and must not be treated as a secret-scalar multiplication path. The timing result is an engineering measurement of the frozen public fixture, not a side-channel assessment.

One-scalar cold panel. The committed manifest, rules, launch, auditor, audit receipt, raw archive, and native provenance preserve serial job a477a7b5198c4a9fa4db825ae0ce03ca on RunPod thyvmdqmzsz5ik, CPU 40. Sixteen alternating pairs, 96 arms, and 48 pairs passed source/binary stability and all A/A controls. Fixture loading, process launch, and output checking are outside Rust Instant; the audited cold interval sums table/selector preparation and evaluation for each arm. The candidate has zero separately reported preparation, so comparing only online_ms would use the wrong boundary.

One frozen scalar, cold	Mean preparation + evaluation	Retained full reference table
Full-table 31+32τ reference	54.023660 ms	495,652 bytes
Scalar-local synthesis	3.240267 ms	Not measured as peak memory

The qualified reference/candidate paired median is 16.736384×, with 95% paired bootstrap interval [16.671217,16.860954] across 16 valid comparisons. The A/A controls passed the predeclared [0.90,1.11] range. This is a Tier 1 one-scalar cold result on one observed allocation; it does not establish batch throughput or a memory advantage.

Full-stream correctness replay. The separate manifest, rules, launch, auditor, receipt, and raw archive preserve job 22a05f971a814d459e3ec804fe967539. Both candidate arms verified all 12,288 outputs; the audit independently replayed the final scalar multiplication and verified the binary/source and input hashes. The arm intervals were 33.874515 s and 33.902518 s. This is a correctness A/A replay, not a matched throughput comparison against the full-table implementation.

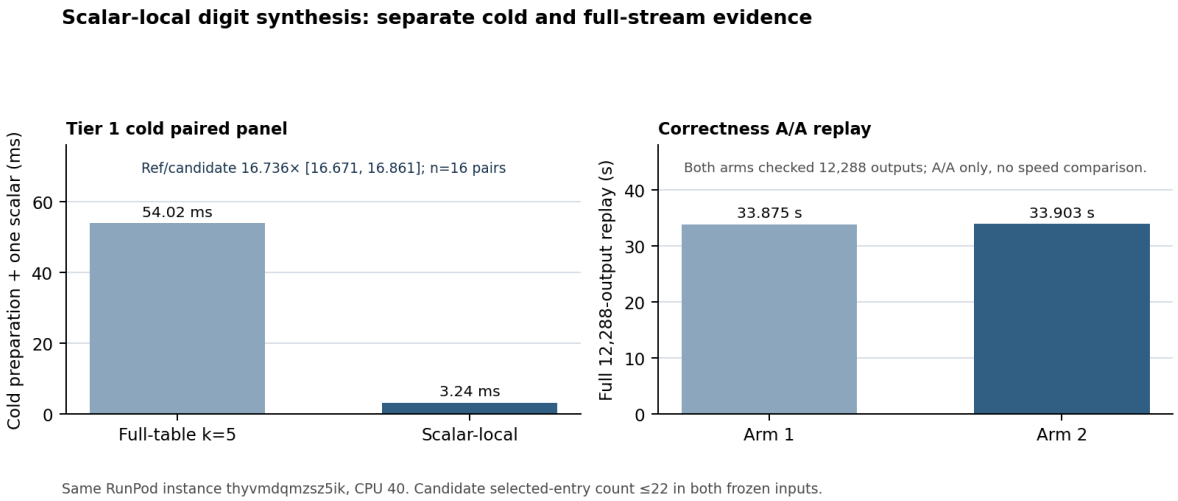


Figure 2: Cold paired panel and distinct full-stream replay

Limits and graph status. The shared sparse-radix proof record tracks the unresolved all-scalar digit-selector and curve group-law bridge. The full replay checks only the committed fixture; it does not prove universal correctness, constant-time behavior, or a peak-memory bound. The follow-on projective builder is a separate implementation path with its own panels. No new curve or map was found. The study has no canonical curve/isogeny graph source to update, and `origin/main` contains no secp256k1 catalog entry. The editable figure source, SVG/180-dpi PNG copies, and this report's PDF accompany the report.